Redpanda featuring aws

Building GenAI apps with a serverless event-driven architecture

Let's get started

Who are we?

Garrett Raska
Solutions Engineer
Redpanda

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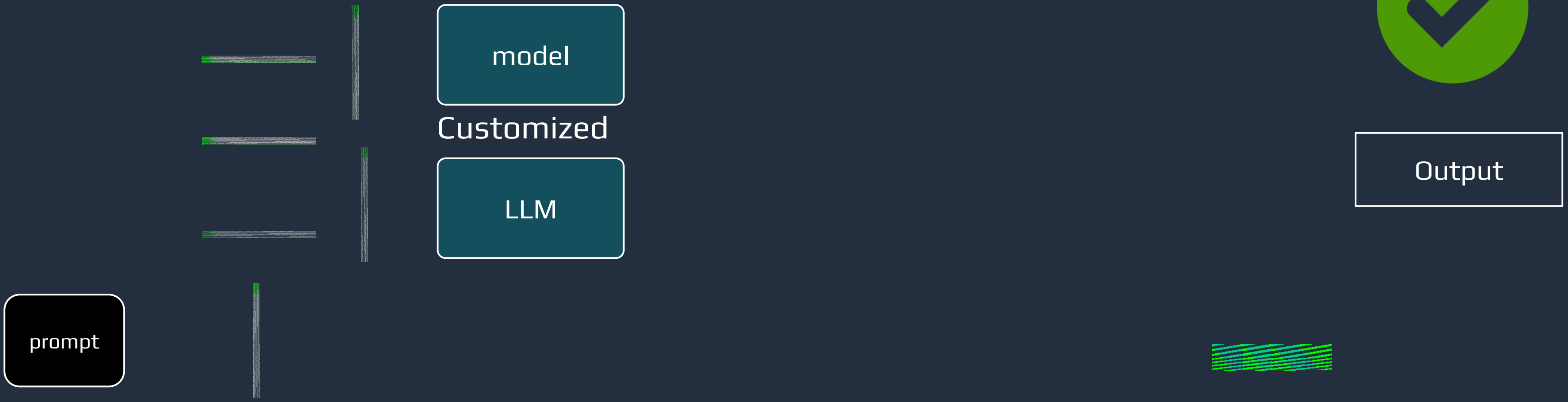
Solutions Architect
AWS

Agenda

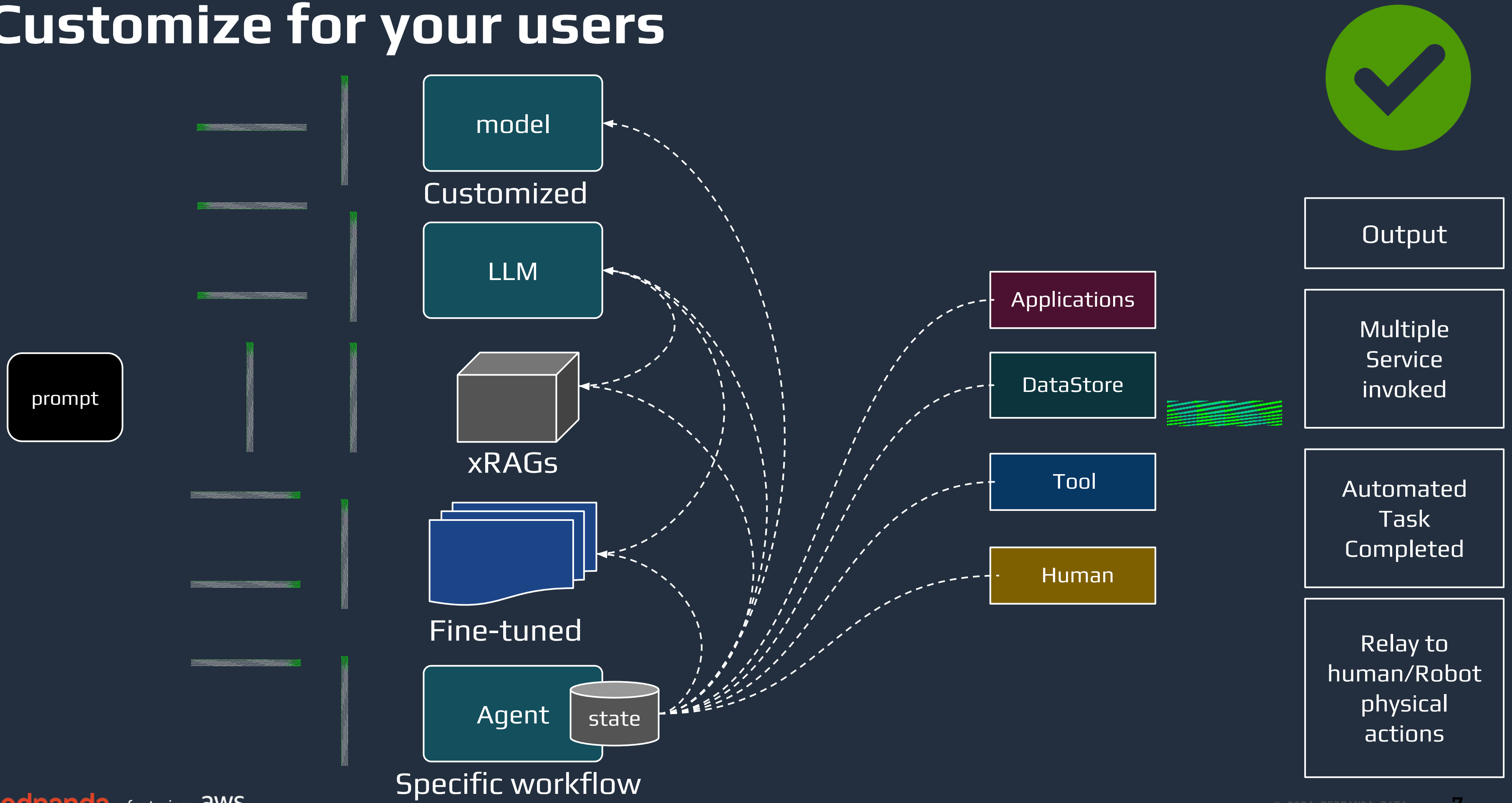
- **Introduction to Serverless EDA** (80 mins)
  - Why serverless and event-driven architecture for GenAI apps (30 mins)
    - Overview of Redpanda streaming, AWS Lambda, and Amazon Bedrock
  - Hands-On Tutorial: Chat with NPCs with Serverless GenAI application (50 mins)
- **Deep Dive into RAG** (40 mins)
  - RAG framework and its significance in enhancing GenAI applications (20 mins)
  - Take home work: Integrating RAG into the game (20 mins)

Why Event Driven & Streaming Data in GenAI?

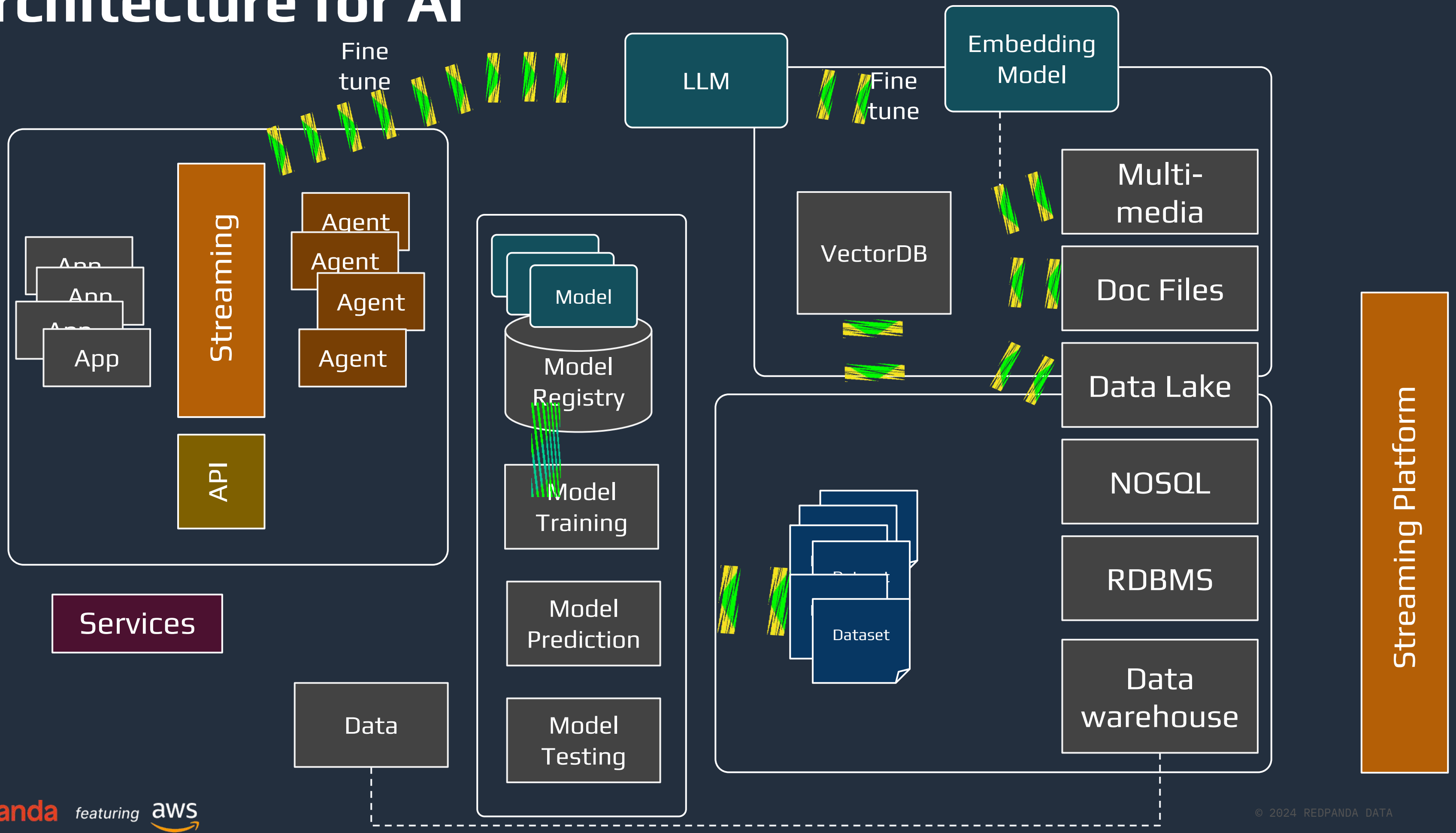
Customize for your users



Customize for your users



Architecture for AI



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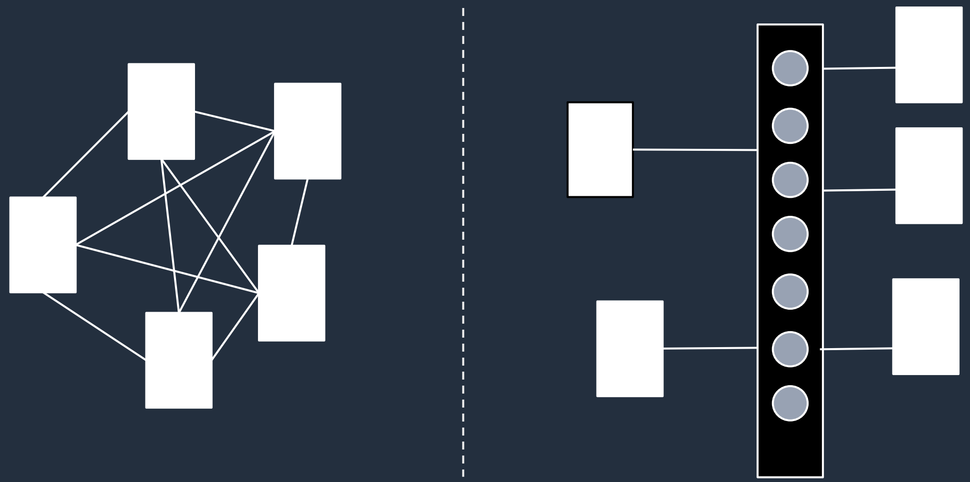
Why Event Driven Architecture?

Event-Driven Architecture (EDA) is a way of designing applications and services to respond to real-time information based on the sending and receiving of information about individual event notifications



Event Driven Architecture

Mirrors the real world

The real world is event-driven. Systems generate and respond to events in everyday life, e.g., the human central nervous system.

Reduced coupling

Traditional RPC-style service architecture results in tightly-bound services. Changes to the application flow typically require service code changes. EDA allows new functionality to be added by adding services that consume existing event streams.

Encapsulation

Microservices concepts have grown in popularity due to the ability for service teams to develop services in isolation. EDA means that service designers need not be aware of how events are consumed.

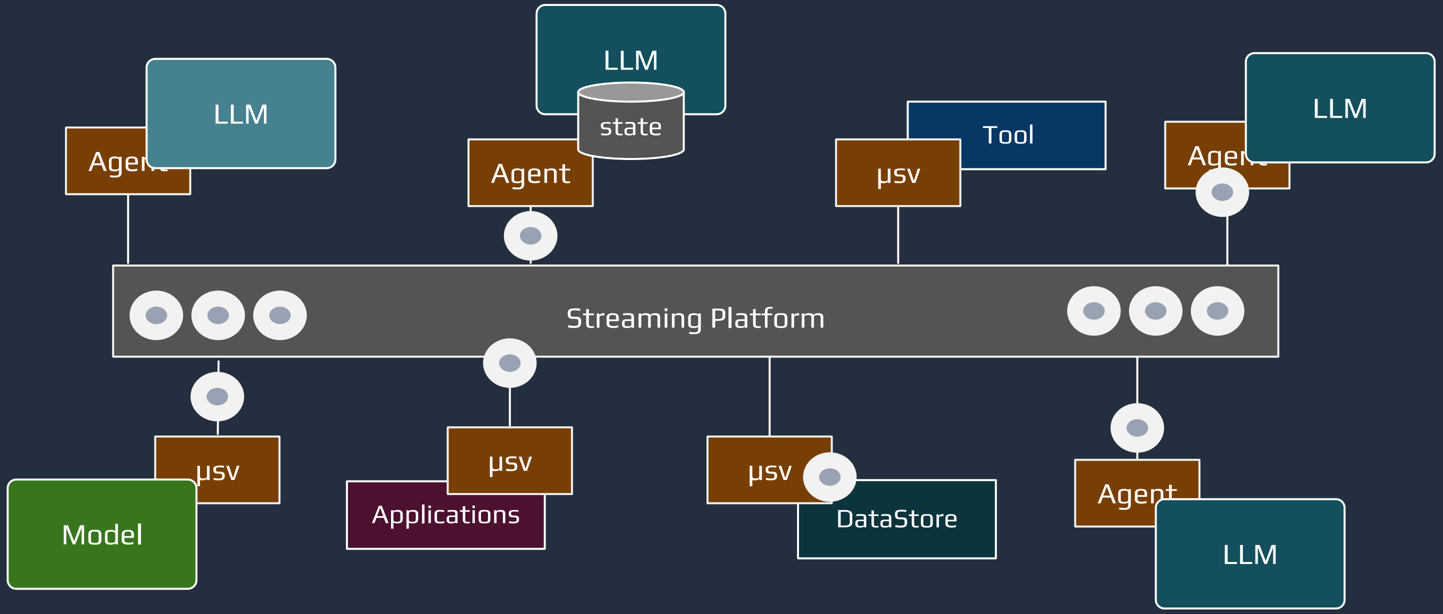
Fine-grained scaling

Services can be independently scaled up and down to meet the event volume.

Near real-time latency

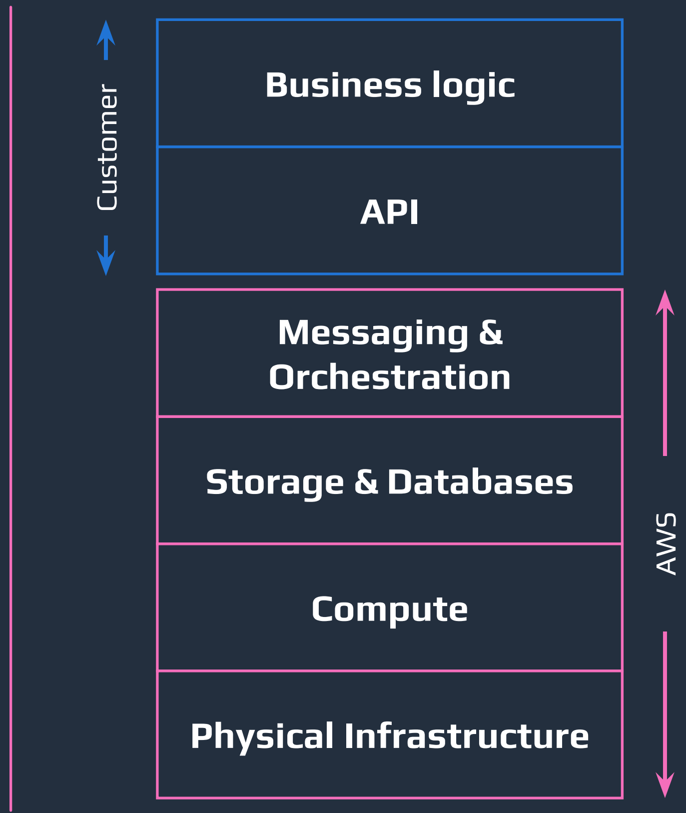
Customers increasing expect a near real-time experience. Polling on APIs is a delicate trade-off between responsiveness and load. EDA allow apps to react in near real-time without compromise.

Streaming and EDA



What is Serverless?

Serverless services simplify the management and scaling of cloud applications by shifting undifferentiated operational tasks to the cloud provider so **development teams can focus on writing code** that solve business problems



Why serverless

TAKE FULL ADVANTAGE OF THE CLOUD TO MODERNIZE APPLICATIONS AND ACCELERATE INNOVATION

No infrastructure provisioning,
no management

Automatic scaling

Pay for value

Highly available and secure

AWS serverless spectrum

AWS OFFERS THE WIDEST PORTFOLIO OF SERVERLESS SERVICES FOR RUNNING AND BUILDING MODERN APPS

From Primitives to…
(Examples: compute, containers, buses)

- AWS Lambda
- AWS Fargate
- Amazon EventBridge
- AWS App Runner
- Amazon ECS
- Amazon SNS
- Amazon S3
- Amazon EFS
- Amazon SQS
- Amazon API Gateway

…Peripherals
(Examples: databases, analytics, workflows)

- Amazon Aurora Serverless
- Amazon DynamoDB
- Amazon Kinesis
- AWS Step Functions
- AWS AppSync
- Amazon CloudWatch
- AWS Glue
- Amazon OpenSearch
- Amazon QuickSight

What is AWS Lambda

Event Source

Changes in data state

Requests to endpoints

Changes in resource state

Function

Business logic

.NET/C#

Node.js, Python, Java, Go,
Ruby, PowerShell
Bring Your Own

Services / Other

Many more…

Benefits of Consuming Stream Data with Lambda

Build faster. Write less code.

With AWS lambda customers do not have to worry about operations like polling the streams for messages. Developers can focus on business logic vs streaming specific tasks. Lambda takes care of polling with built-in pollers. No need to use any stream specific libraries or APIs to process/poll the streams.

Reduce operational overhead

No need to worry about infrastructure management tasks like capacity provisioning and patching.

Automatic Scaling

Scale up or down as needed from a few requests per day to thousands per second and ensure consistently low latency.

Security and built in availability

Lambda runs your code on a high-availability compute infrastructure with security isolation

Lower TCO for variable workloads

Reduce costs by paying only for the compute time you use. Only run when there are events to process.

Announcing Provisioned Mode for Kafka ESM

NEW

Configurable **minimum** and **maximum** number of **always-on** event pollers

Faster scaling great **for latency-sensitive** workloads

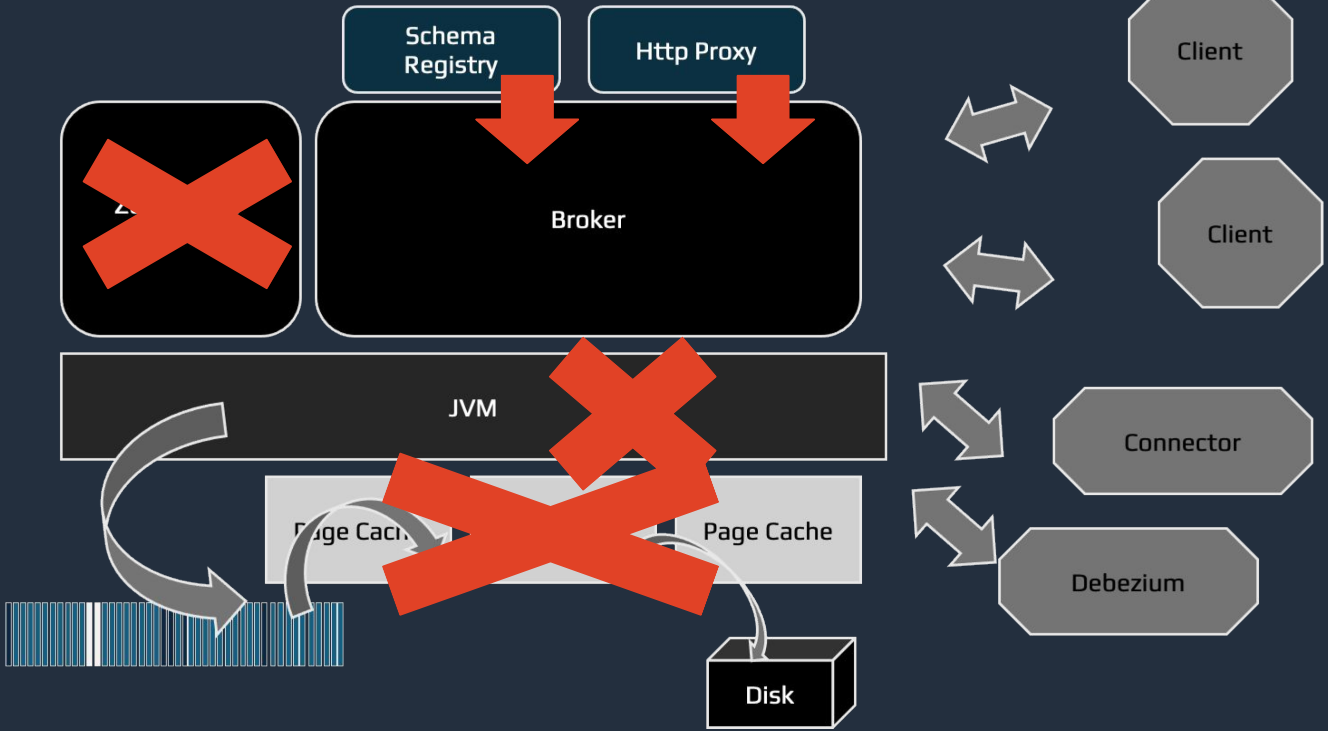
Introducing Amazon Bedrock

The easiest way to build and scale generative AI applications with foundation models (FMs)

- Choice of leading FMs through a single API
- Model customization
- Retrieval Augmented Generation (RAG)
- Agents that execute multistep tasks
- Security, privacy, and safety

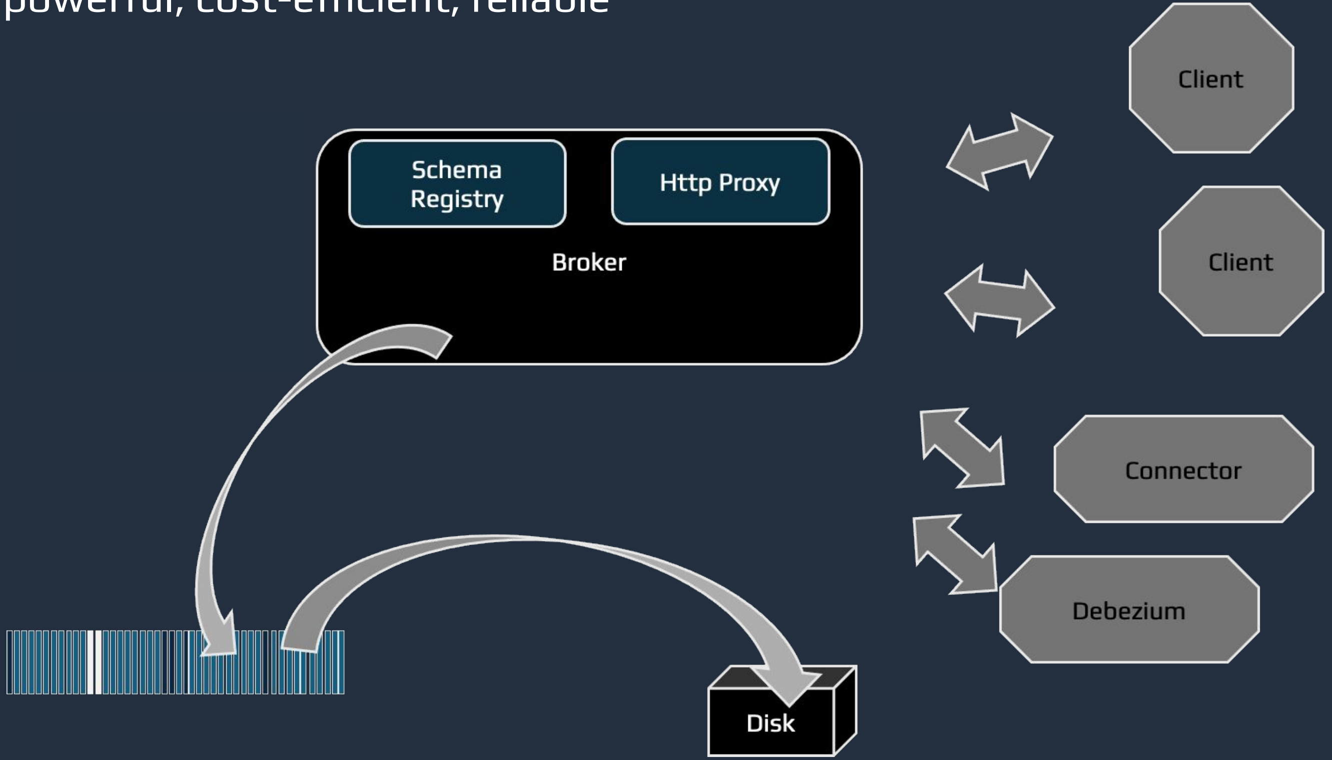
Redpanda

Simple, powerful, cost-efficient, reliable



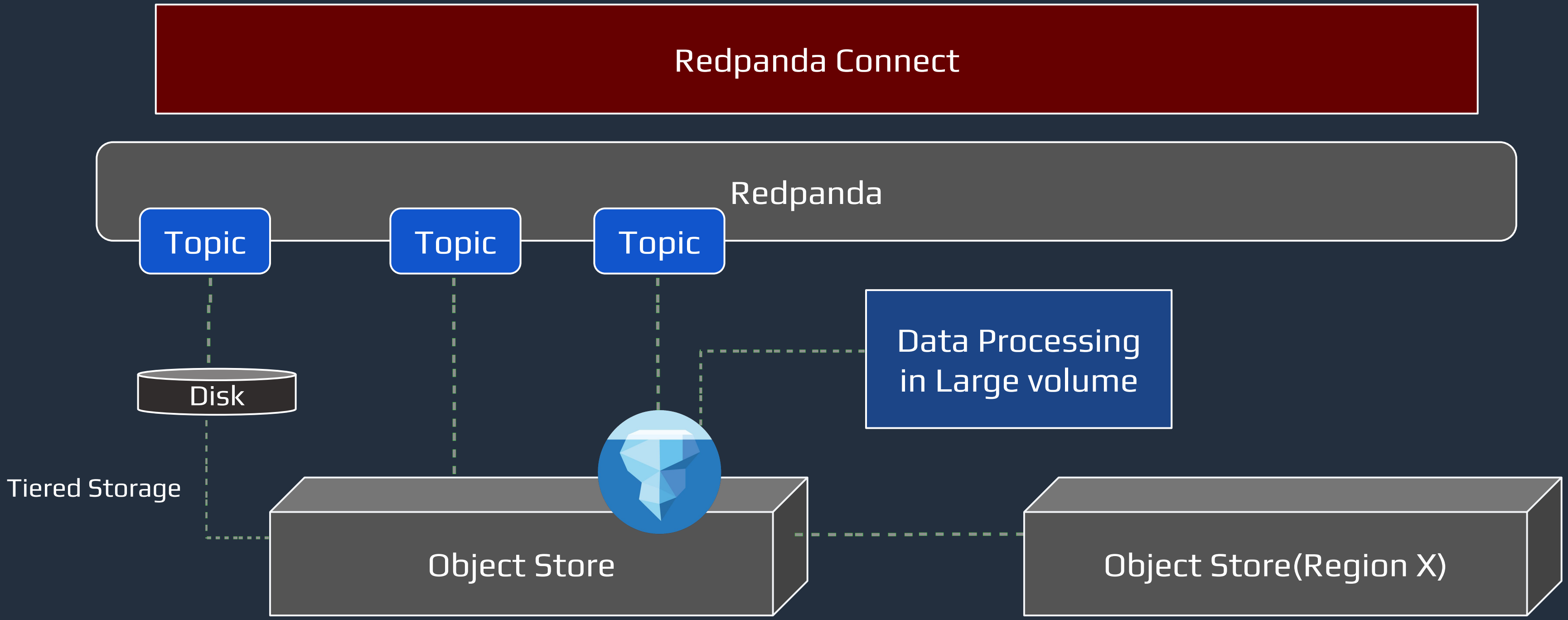
Redpanda

Simple, powerful, cost-efficient, reliable

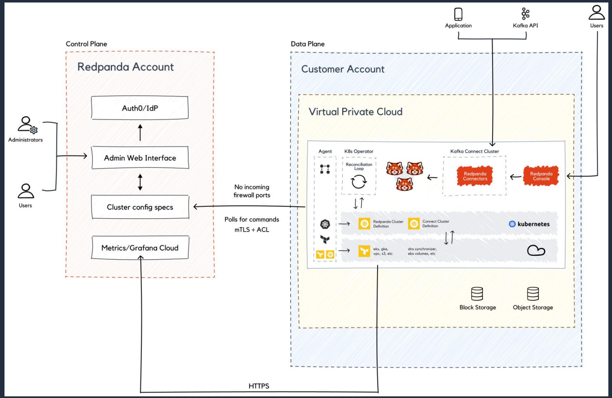


Enter Redpanda

Redpanda One

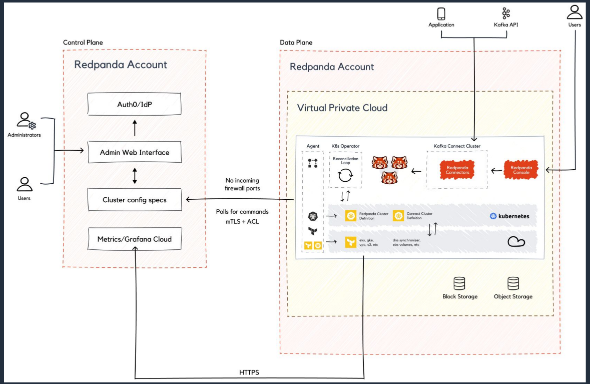


Your choice: BYOC & Dedicated clusters & Serverless



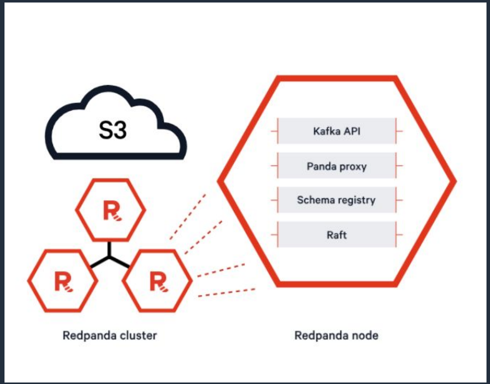
BYOC Clusters

Data plane runs in customer's account. Full data sovereignty and ability to leverage your cloud credits!



Dedicated Clusters

Everything runs in Redpanda's account. Configurable instance sizes with complete isolation from other workloads.



Serverless Clusters

Fully managed, instant available Redpanda managed cluster

Redpanda

Why build Gen AI on AWS Serverless Compute?

Combining Speed and Power

Speed of Serverless

+

Power of Gen AI

=

Rapid delivery of smarter applications and features with focus on Innovation

AWS Lambda

Amazon ECS

AWS Fargate

AWS Step Functions

Amazon EventBridge

Amazon SageMaker

Amazon Bedrock

Amazon Q

Why Serverless Compute for generative AI?

- Focus on AI Code
- Flexible Integrations
- Cost-effective
- Agility with rapidly evolving FMs

Why Serverless Compute for generative AI?

- **Focus on AI Code**
- Flexible Integrations
- Cost-effective
- Agility with rapidly evolving FMs

Built-in auto scaling, high availability, and fault tolerance to ensure developer productivity **alleviating teams from the complexity of infrastructure planning and management** for high throughput **inference workloads**.

Why Serverless Compute for generative AI?

Focus on AI Code

Flexible Integrations

Cost-effective

Agility with rapidly evolving FMs

Quickly integrate and deploy FMs into your applications and workloads running on AWS **using familiar controls and integrations with the depth and breadth of AWS capabilities and services** like Amazon SageMaker and Amazon Bedrock.

aws

Why Serverless Compute for generative AI?

Focus on Code

Flexible Integrations

Cost-effective

Agility with rapidly evolving FMs

Cost-effectively scale infrastructure to train and run FMs containing hundreds of billions of parameters by **pay-by-request** by paying for the duration and quantity of inferences.

Why Serverless Compute for generative AI?

Focus on Code

Flexible Integrations

Cost-effective

Agility with rapidly evolving FMs

Generative AI is in a very early stage of technology, **build a company culture for continuous experimentation** and evaluation of different models. Build with serverless architectures like EDA etc by reducing the interdependency where teams can work independently to develop, deploy and scale their Gen AI solutions.

Langchain

Chain

- Brings together components to create meaningful responses from language models.

User Input

Prompt

- The entry point for interacting with LLMs, directing the flow of information.

Model

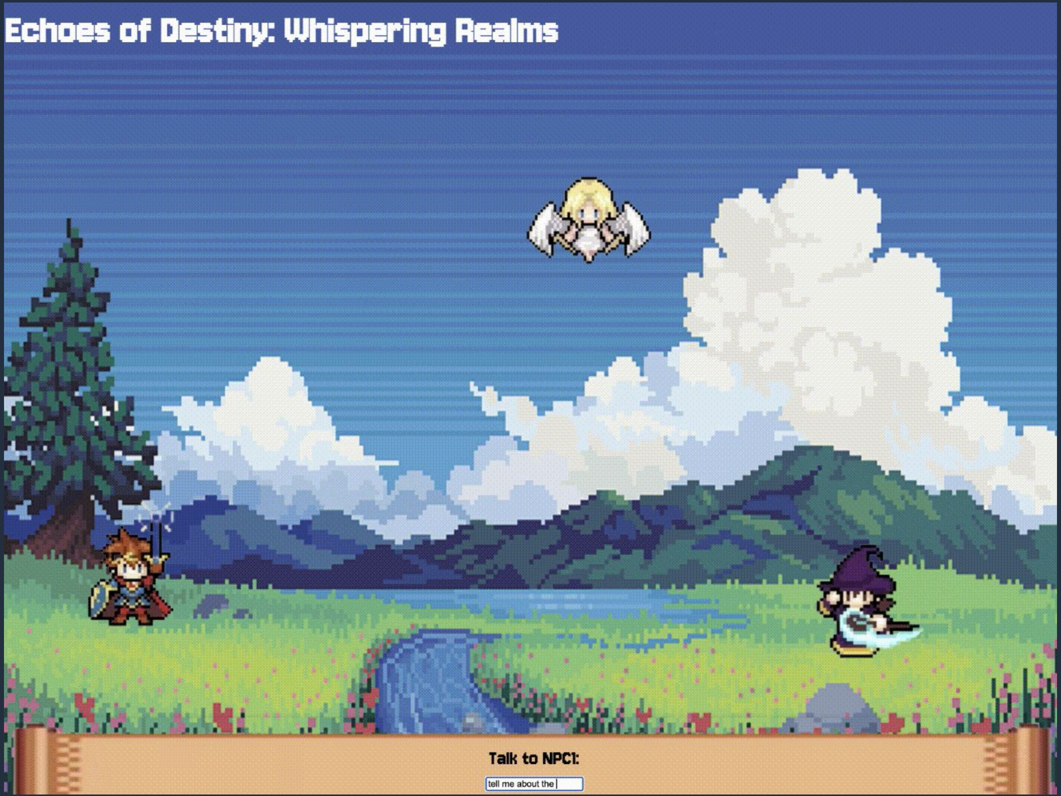
- The core of LangChain, consisting of three primary types: Large Language Models (LLMs), Chat Models, and Text Embedding Models.

Agent

- Brings together components to create meaningful responses from language models.

Workshop

Overview



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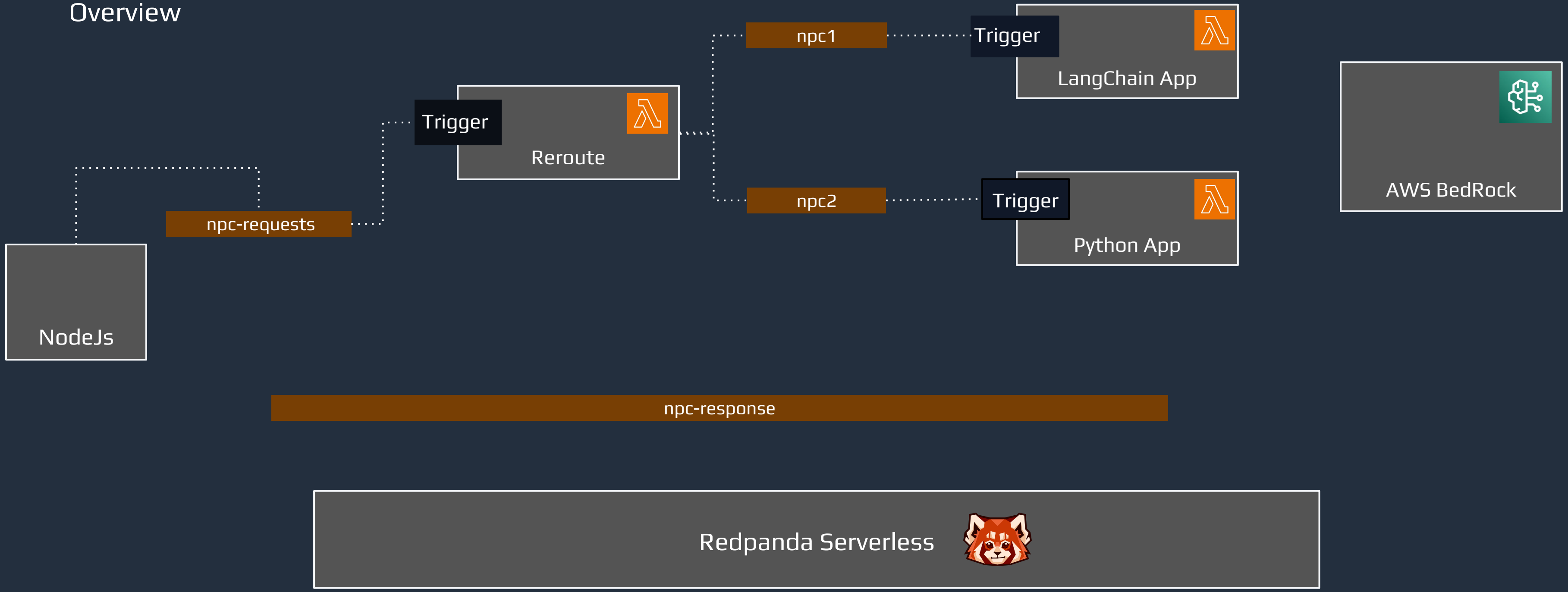
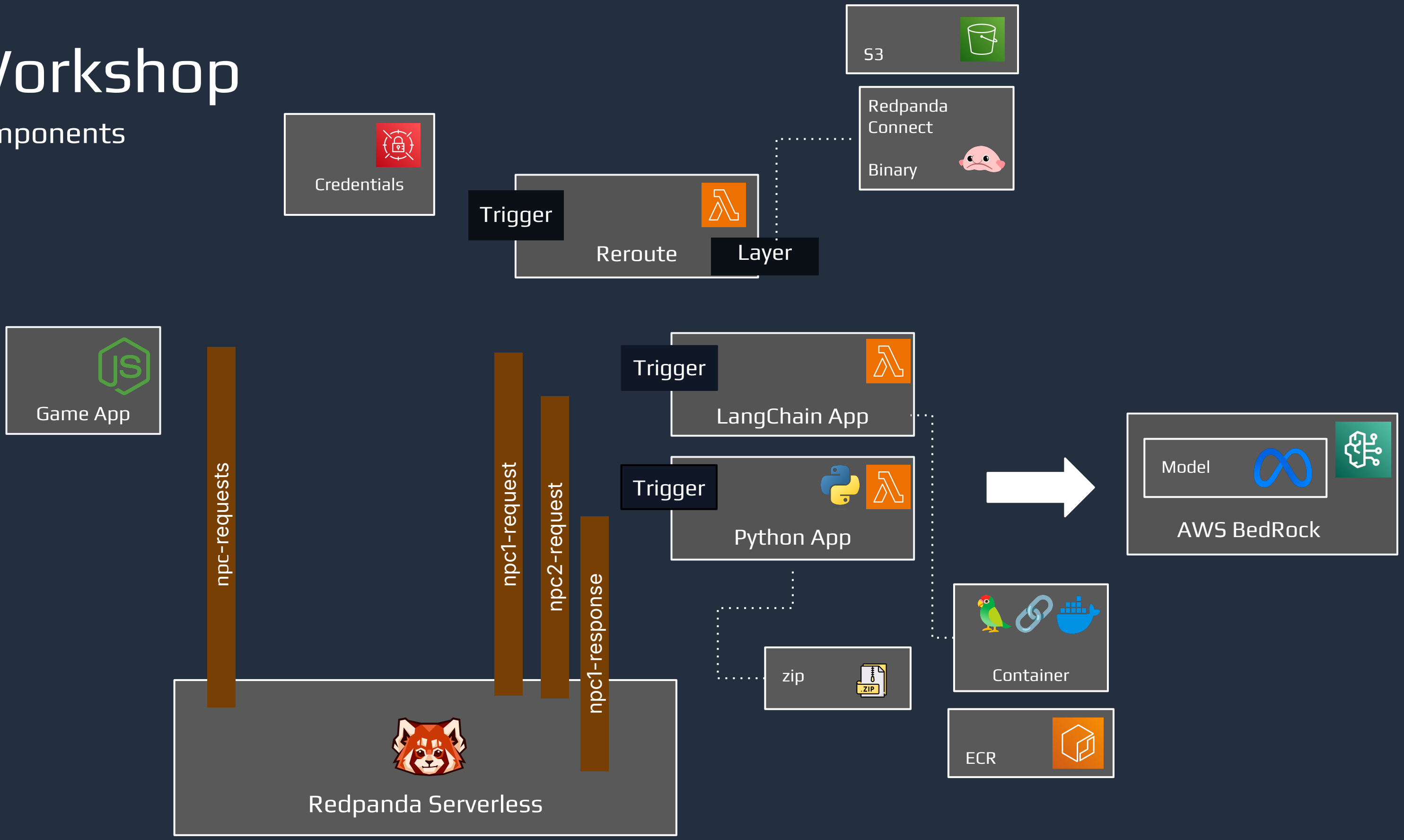
Workshop

Overview



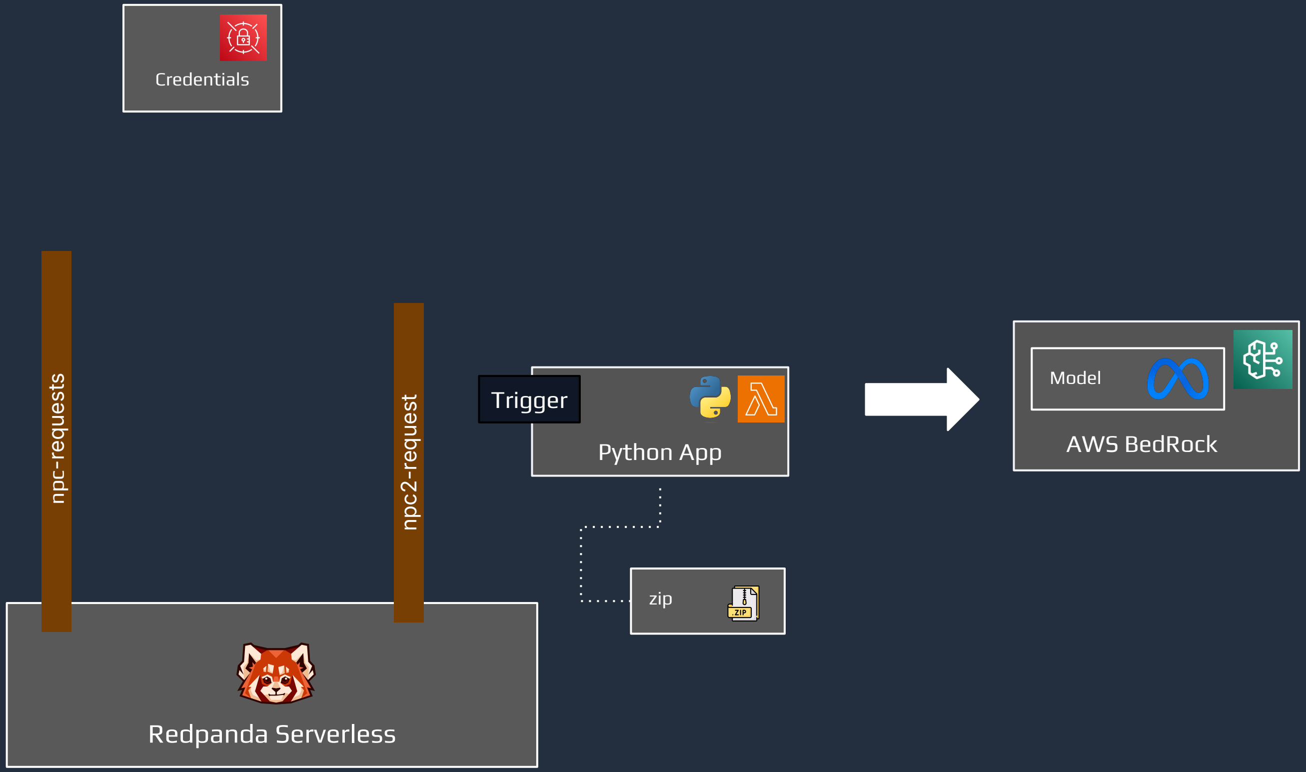
Workshop

Components



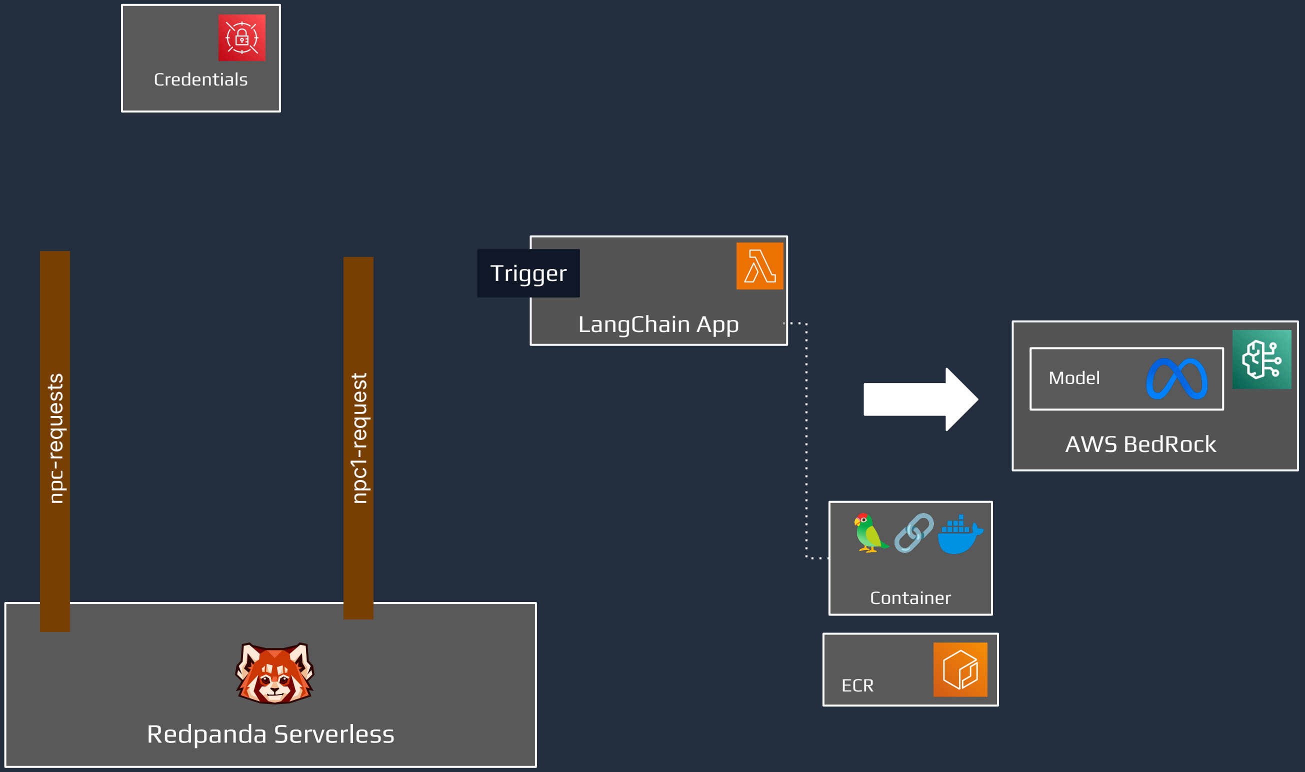
Workshop

Components



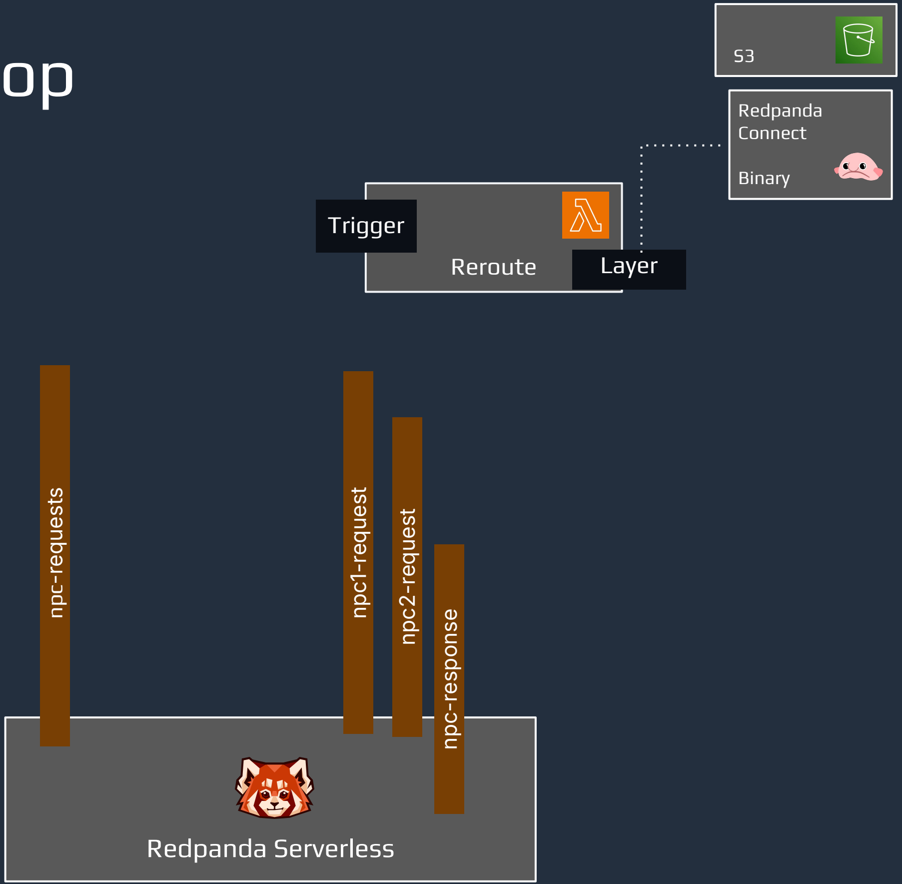
Workshop

Components



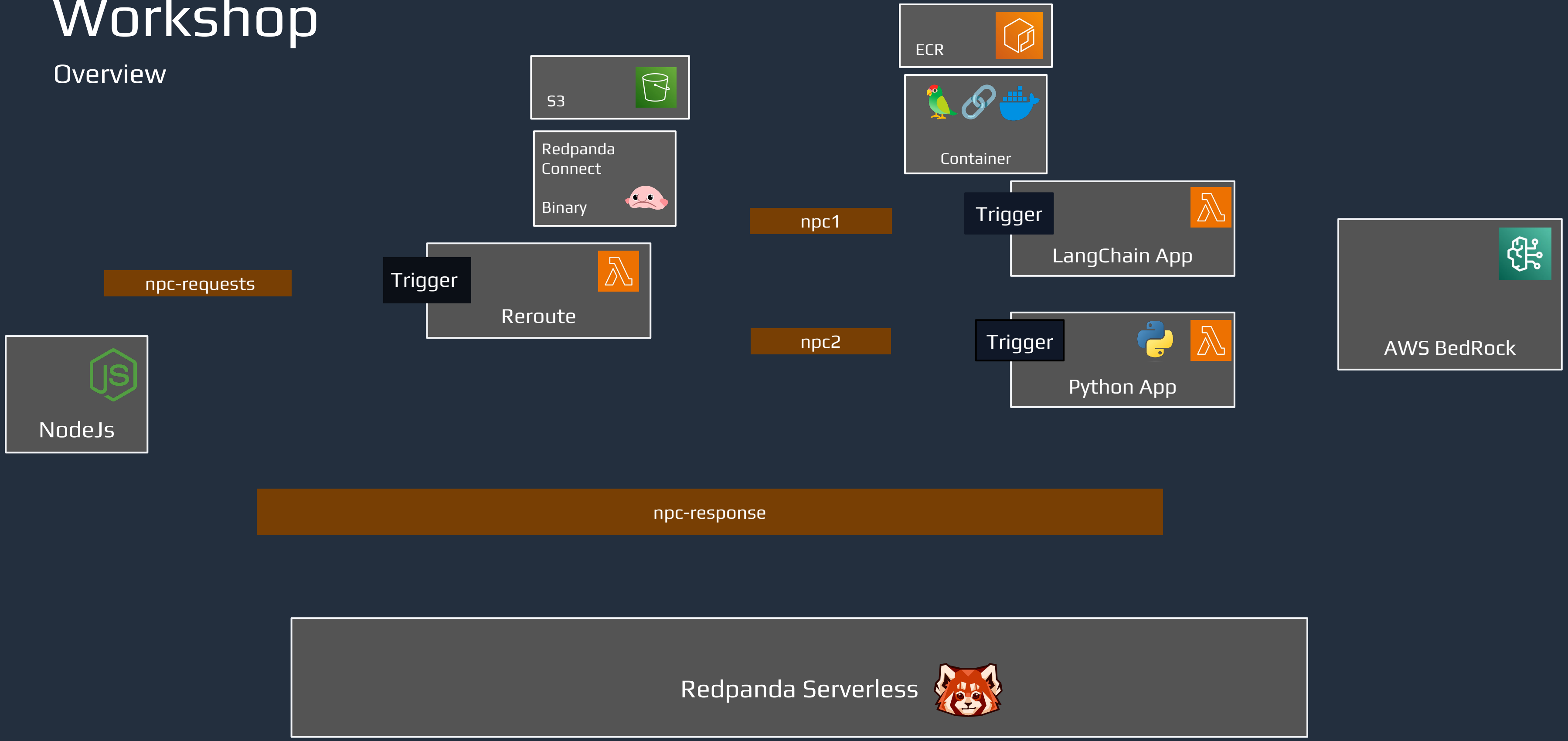
Workshop

Components



Workshop

Overview



Workshop

https://catalog.us-east-1.prod.workshops.aws/join?access-code=fe61-0c9dbc-34

https://play.instruqt.com/redpanda/invite/ywkbm7urygma

Introduce RAG

Challenges

Depends heavily on the quality and quantity of data and robustness of the algorithms. LLMs may struggle to differentiate between accurate data and misinformation, lead to factually incorrect or contextually inappropriate output.

Cost

Inaccuracy

Data Security

IP Protection

Challenges

The need to expose critical private information to third-party vendors for data training. This exposure increases the risk of sensitive data being mishandled or misused, leading to potential data breaches. Public AI services over networks adds another layer of risk, as data in transit can be vulnerable to interception or unauthorized access.

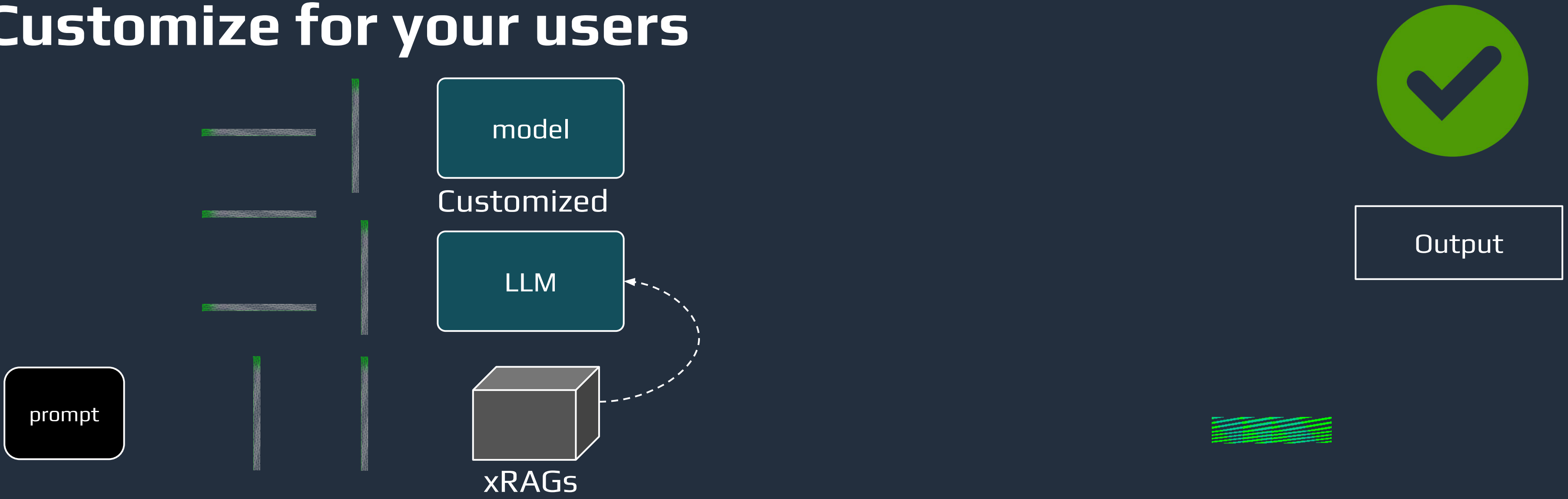
Cost

Inaccuracy

Data Security

IP Protection

Customize for your users



xRAGs

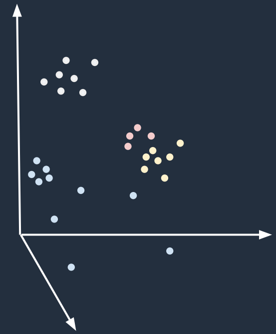
Prompt

Schema

Rules

Background

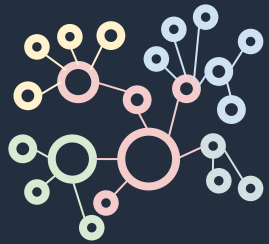
RAG



Additional Information
—
Semantic Search
Similarity Search

GraphRAG

(According to MicroSoft)



Additional Information
—
Relationships
Patterns
Decision optimization

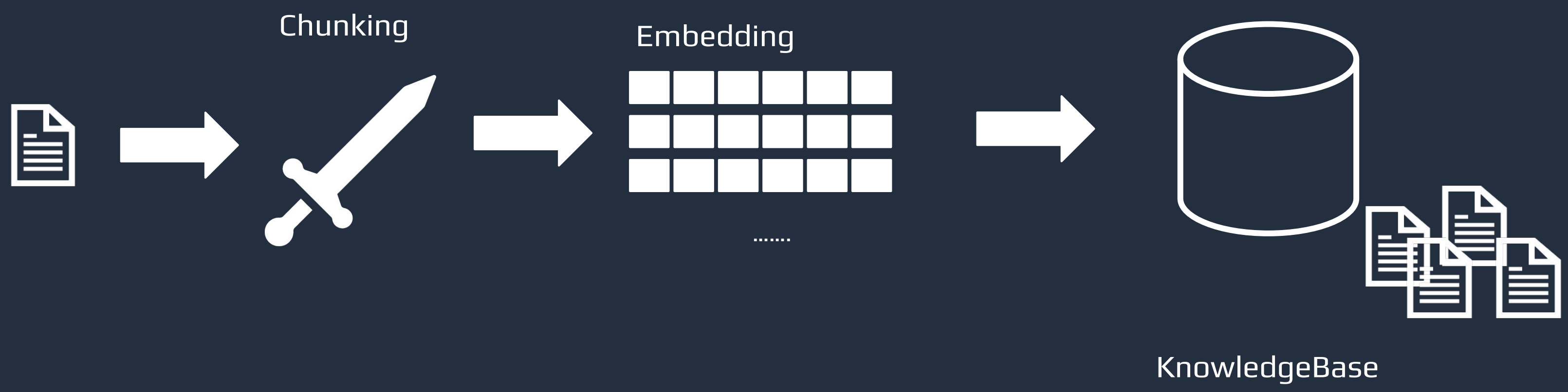
LLM

MemoRAG

Memory

Additional Information
—
Chat History
Interaction summary
Data summary
Feedback

RAG



Embedding

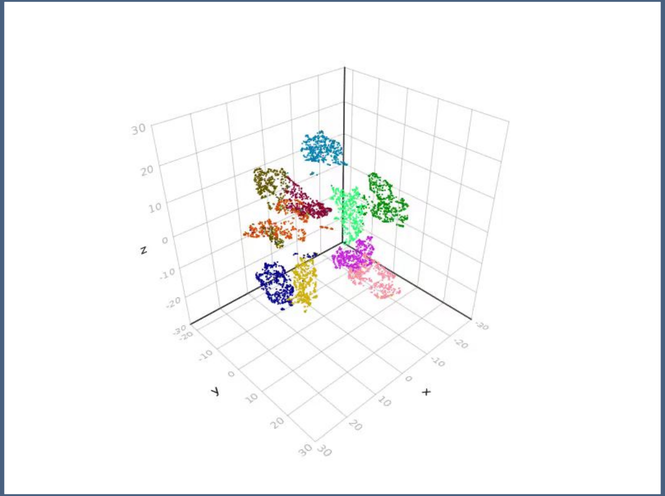
A technique used to represent high-dimensional data in a lower-dimensional space.

"The hero just defeated a monster"

[-4.4954e-01, -5.0617e-01, -1.3257e-01, 2.9100e-01, 8.7137e-03,
-1.6872e-02, 1.7906e-01, 3.0540e-01, -3.5011e-02, -3.0029e-01,
-8.5384e-02, -2.4782e-01, 1.3777e-01, 5.8335e-01, -3.2837e-01,
...]

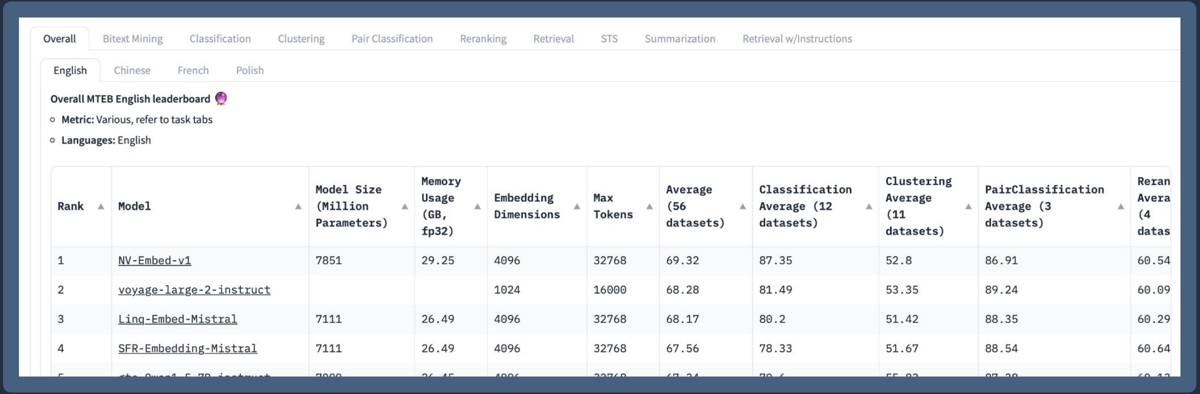
"The battle was fierce, with heo clashing swords with the King monster"

[9.7624e-02, -2.8325e-01, -1.0779e-01, 6.2398e-02, -6.0403e-01,
-1.3933e-01, -1.0179e-01, 3.6617e-01, 7.8573e-02, -8.6283e-02,
1.0246e-02, -2.3709e-01, 3.1932e-01, 4.6958e-01, 1.1292e-01,
-1.5850e-01, -3.3792e-02, -1.0489e-01, -1.0260e-01, 5.3405e-02,
1.0568e-01, -1.4227e-01, -4.2821e-03, 7.7303e-01, 5.8477e-01,
...]



Embedding

Massive Text Embedding Benchmark (MTEB) Leaderboard



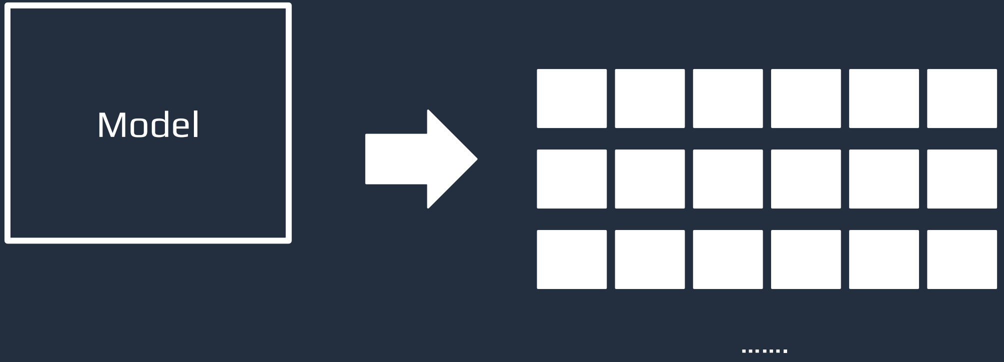
Overall | Bitext Mining | Classification | Clustering | Pair Classification | Reranking | Retrieval | STS | Summarization | Retrieval w/Instructions

English | Chinese | French | Polish

Overall MTEB English leaderboard

- **Metric:** Various, refer to task tabs
- **Languages:** English

Rank	Model	Model Size (Million Parameters)	Memory Usage (GB, fp32)	Embedding Dimensions	Max Tokens	Average (56 datasets)	Classification Average (12 datasets)	Clustering Average (11 datasets)	PairClassification Average (3 datasets)	Reran Avera (4 datas
1	NV-Embed-v1	7851	29.25	4096	32768	69.32	87.35	52.8	86.91	60.54
2	voyage-large-2-instruct			1024	16000	68.28	81.49	53.35	89.24	60.09
3	Linq-Embed-Mistral	7111	26.49	4096	32768	68.17	80.2	51.42	88.35	60.29
4	SFR-Embedding-Mistral	7111	26.49	4096	32768	67.56	78.33	51.67	88.54	60.64

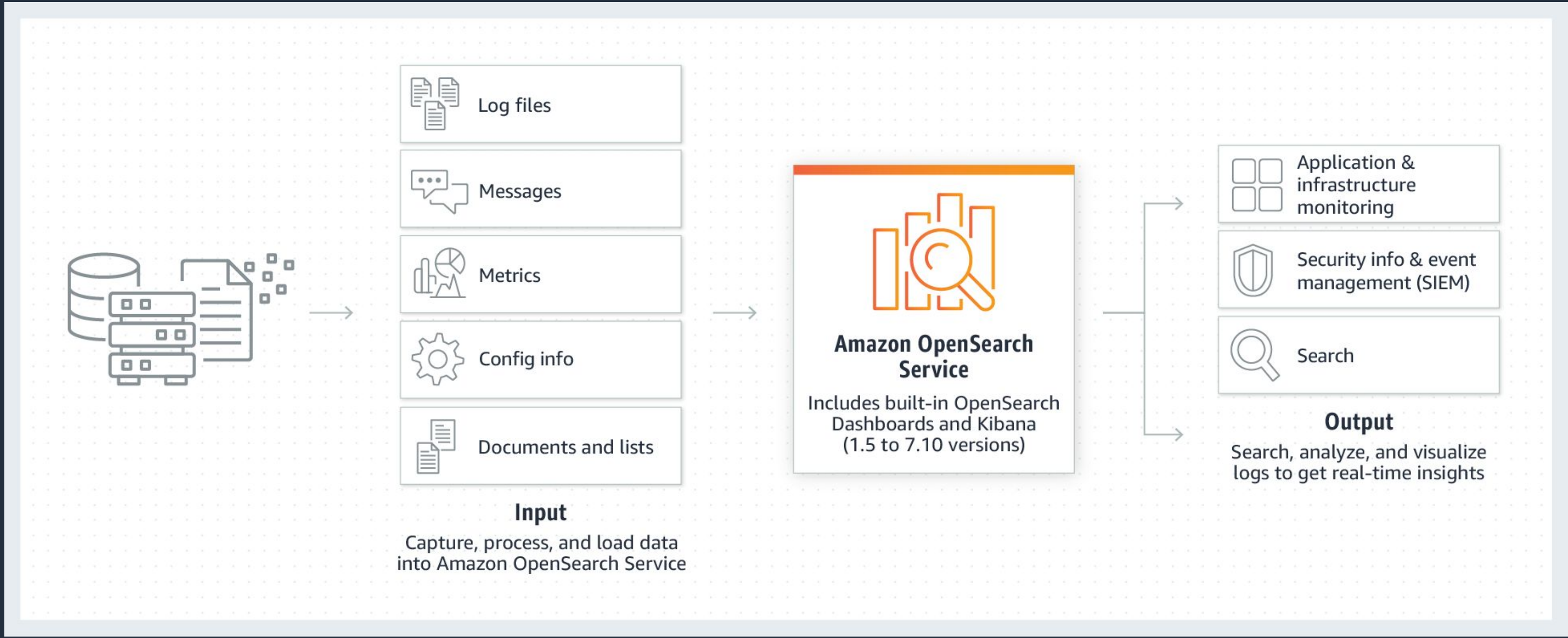


Vetor Database

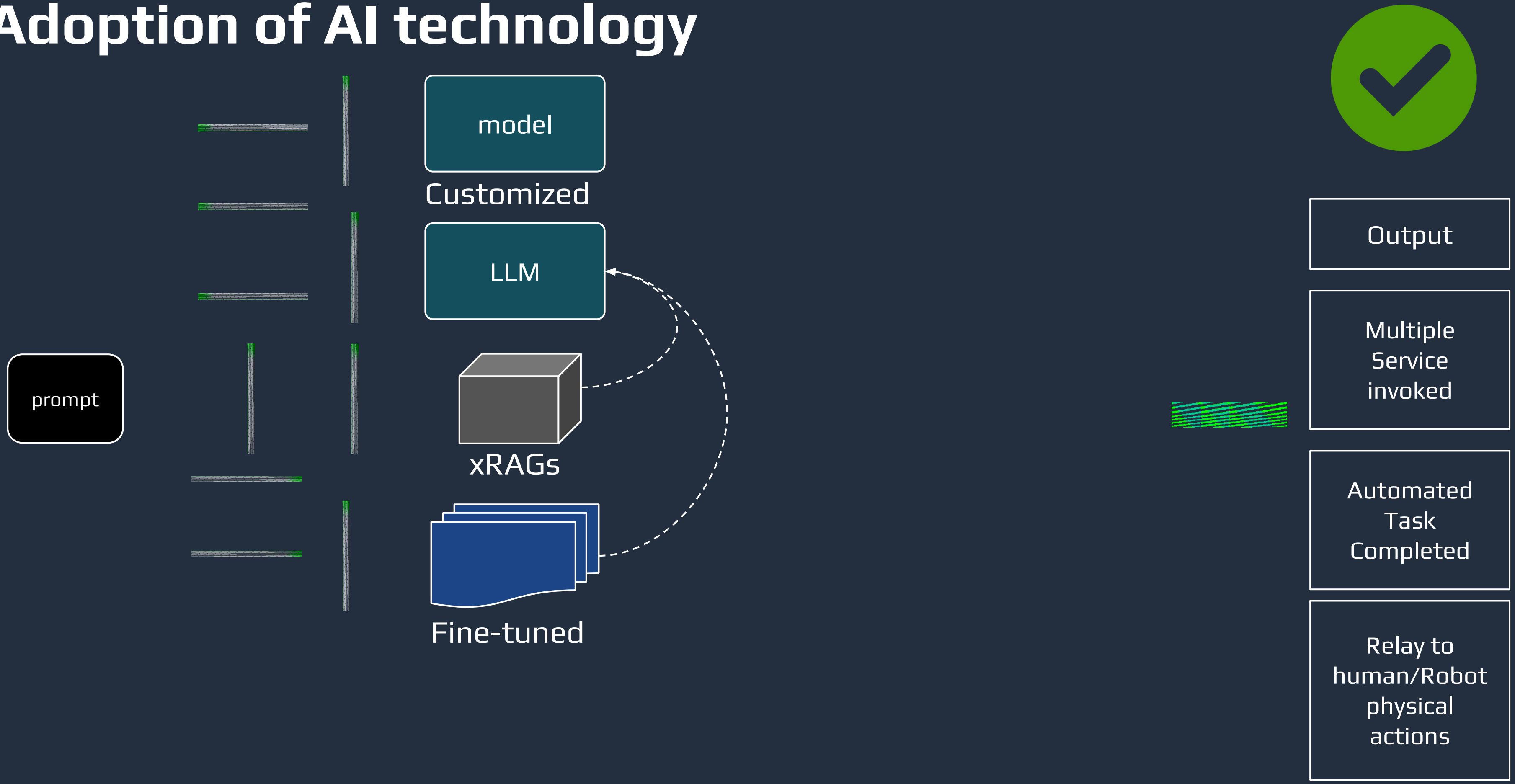
A specialized database designed to efficiently store, index, and query high-dimensional vectors.

- **Scalability and adaptability**
- **User-friendly interfaces**
- **Security data privacy**

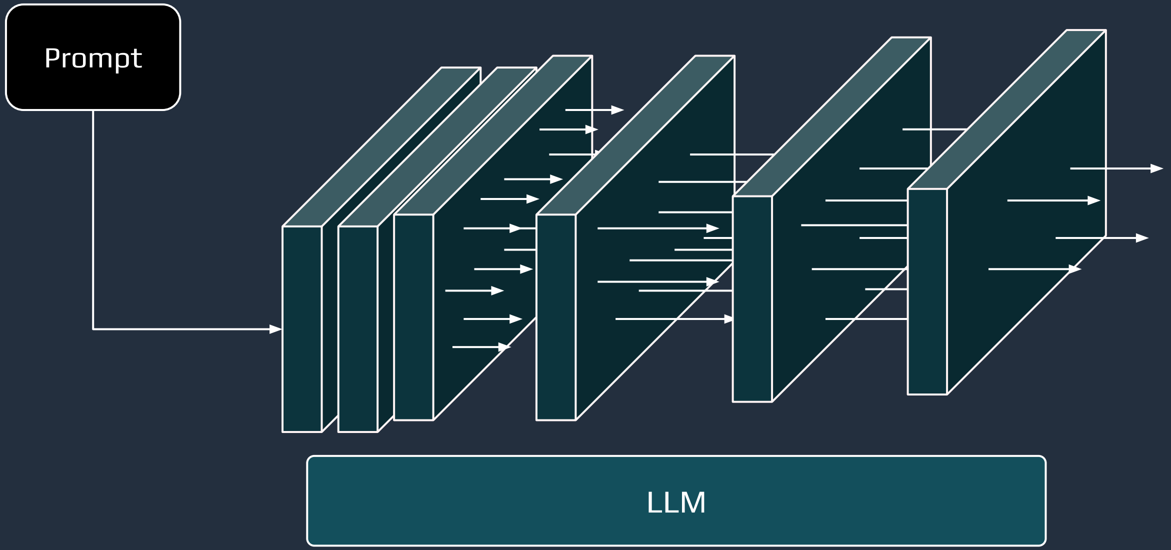
Opensearch Serverless



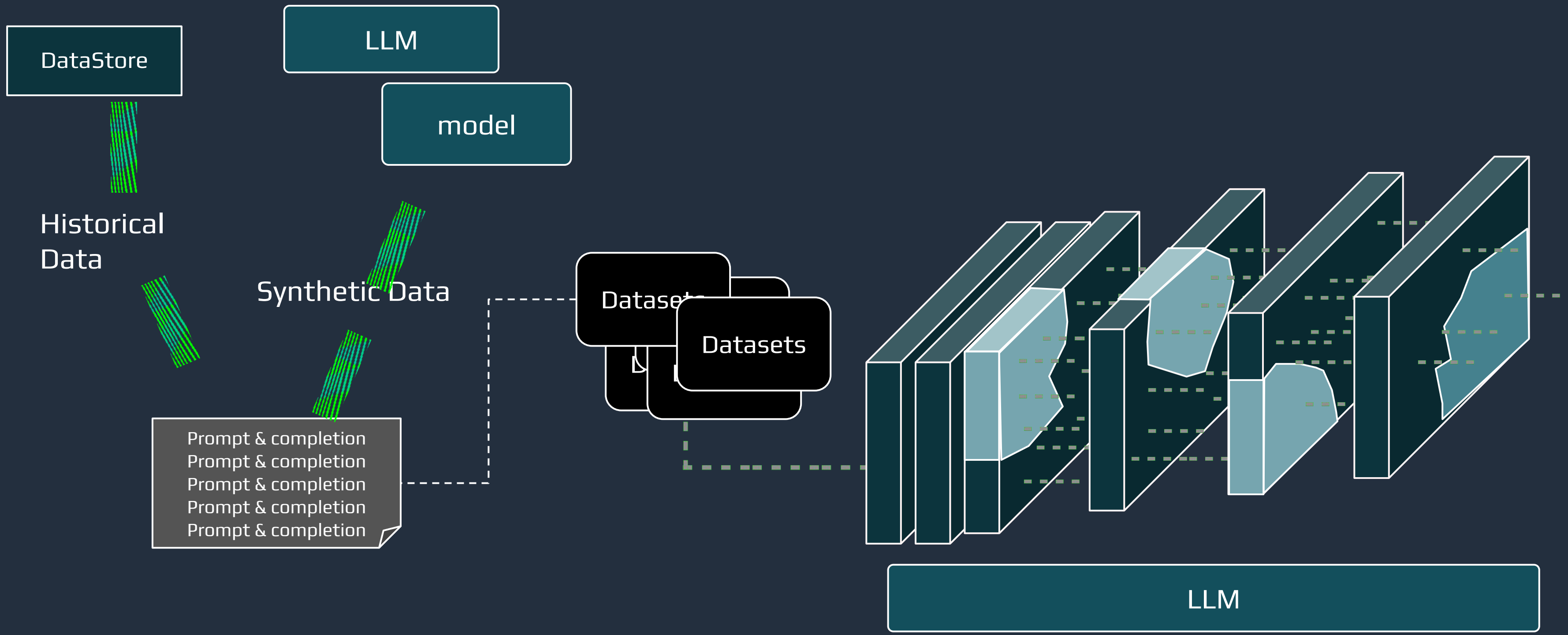
Adoption of AI technology



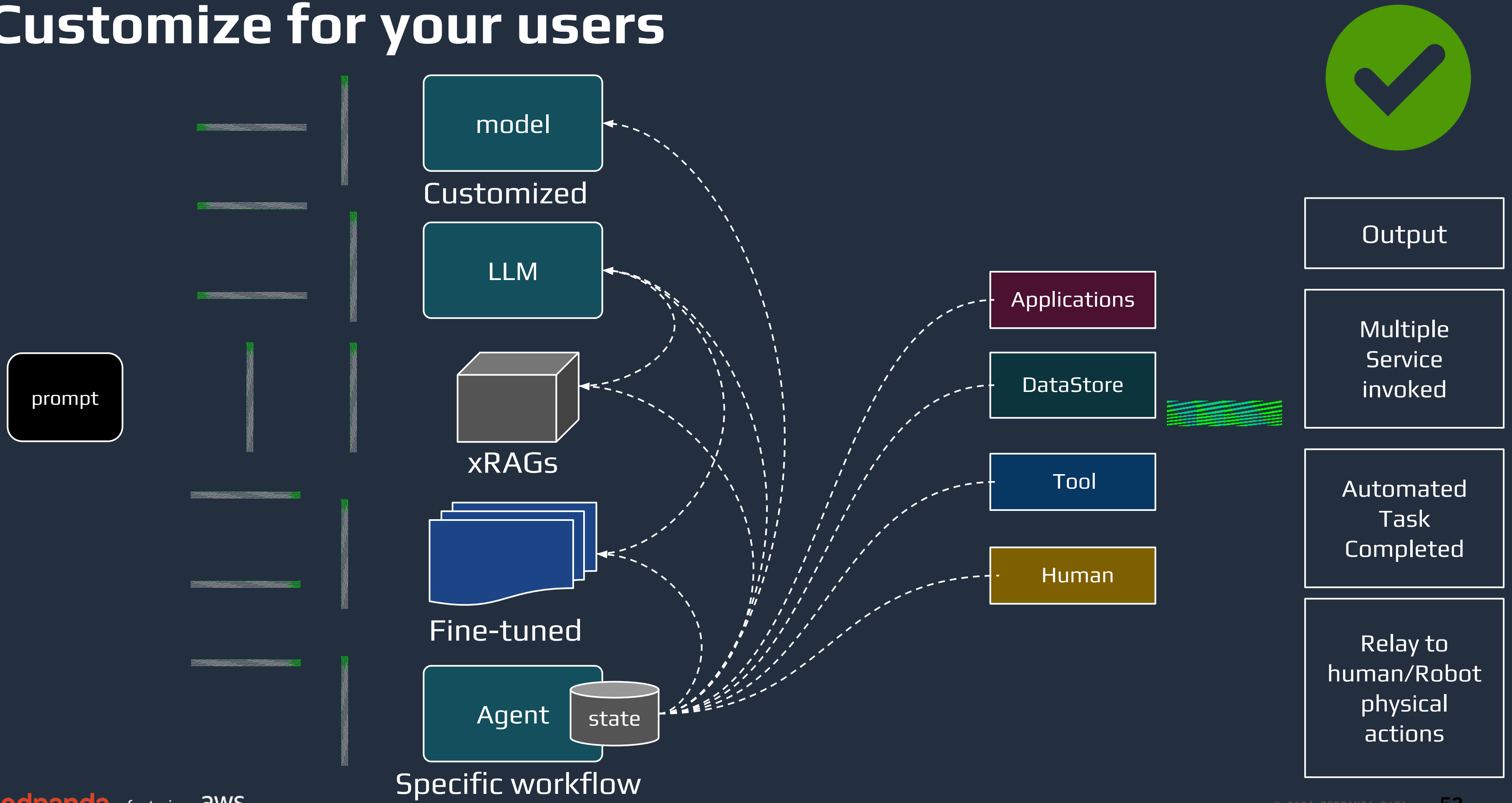
Fine tuning



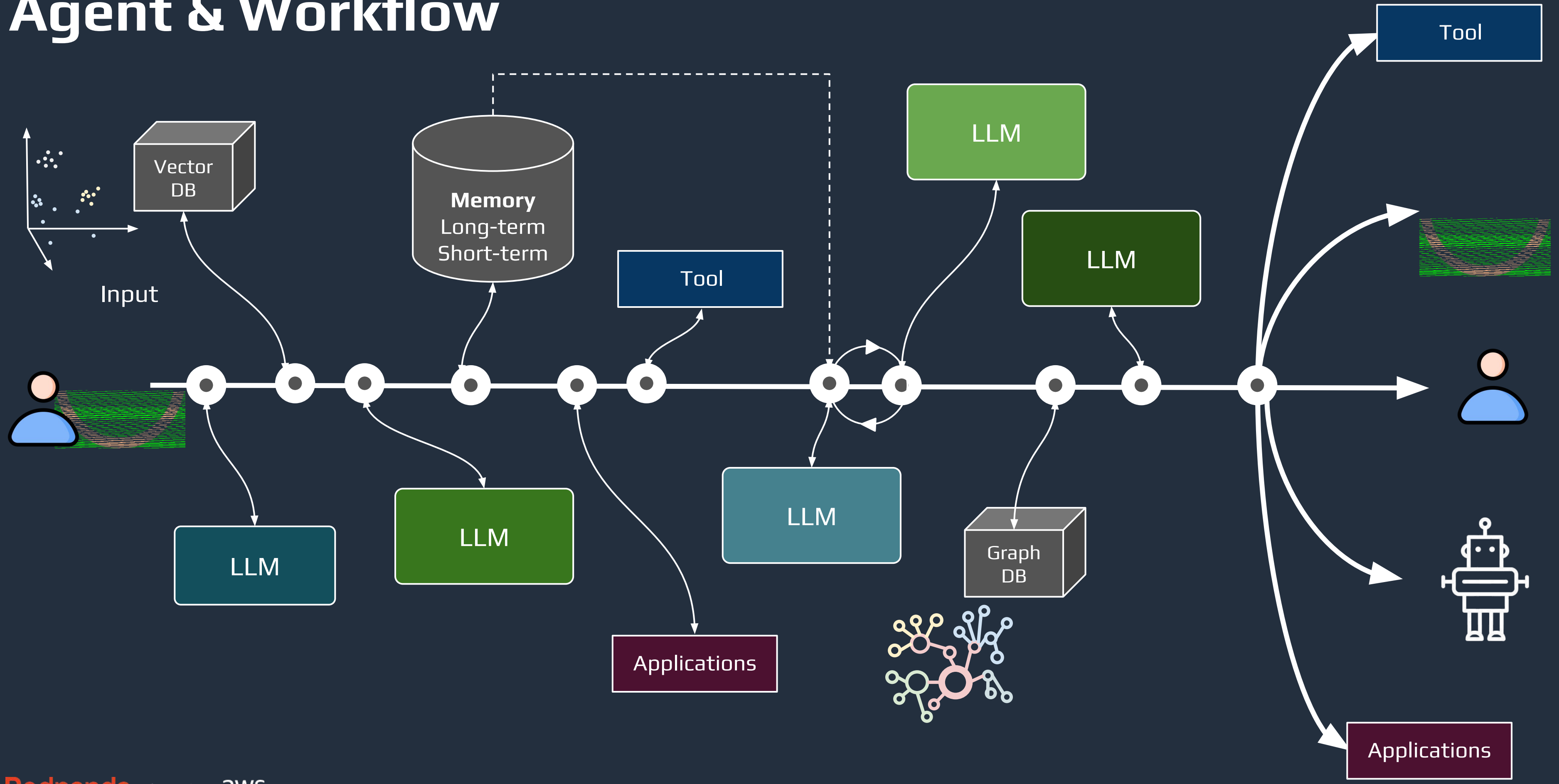
Fine tuning


DataStore
LLM
model
Historical Data
Synthetic Data
Datasets
Prompt & completion
Prompt & completion
Prompt & completion
Prompt & completion
Prompt & completion
LLM

Customize for your users

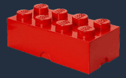


Agent & Workflow



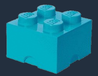
Redpanda Connect

Input



```
input:
  sftp:
    address: 123.45.678.90
    paths: ["upload/*.csv"]
    codec: lines
```

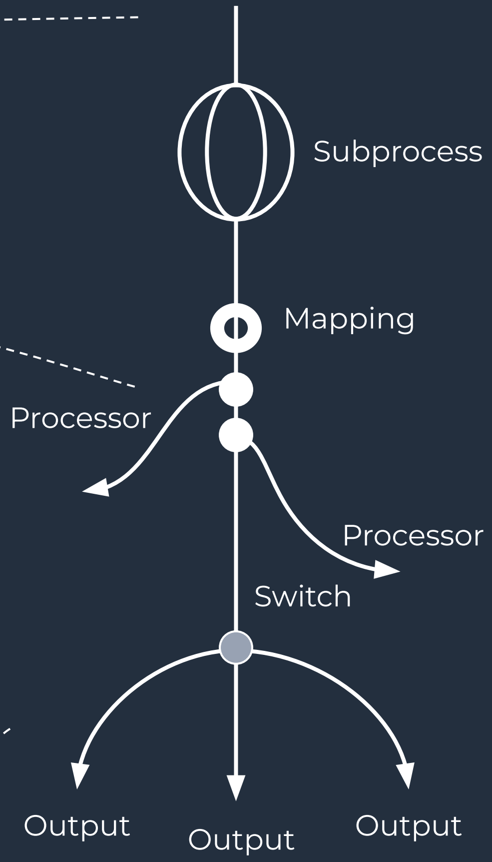
Processor



```
pipeline:
  processors:
    - bloblang: |
        root.id = this.msg_id
        root.body = this.content
    - branch:
         processors:
           -aws_lambda:
                 function: calc_interest
         resultmap: root.meta,interest = meta
```

Output

```
output:
  gcp_pubsub:
    project: "demo"
    topic: "assets"
    max_in_flight: 64
```



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Intuitive Declarative Configuration

Error handling
- retries
- drop
- reject
- DLQ

Bloblang
- jq and jsonpath
- commonly used functions
- frequently used methods
- advance mapping features

Buffer
- batching
- system window

Cache
- deduplication
- data joins
- enrichment

Observability
- metrics
- tracing
- healthchecks

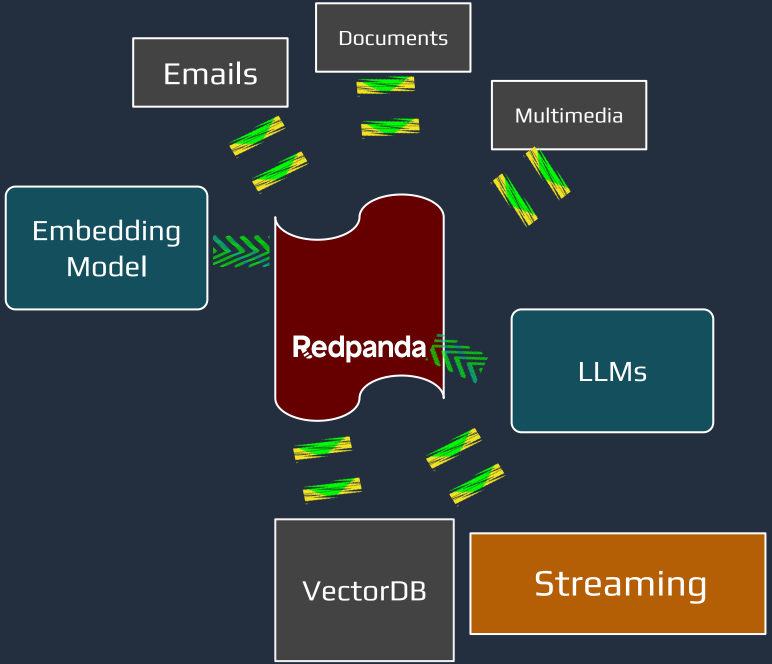
Input

Processor

Output

Redpanda Connect

Piping unstructured data



Input

```yaml
input:
  file:
    paths:
```

Processor

```yaml
processors:
    - ollama_embeddings:
      model: nomic-embed-text
```

Processor

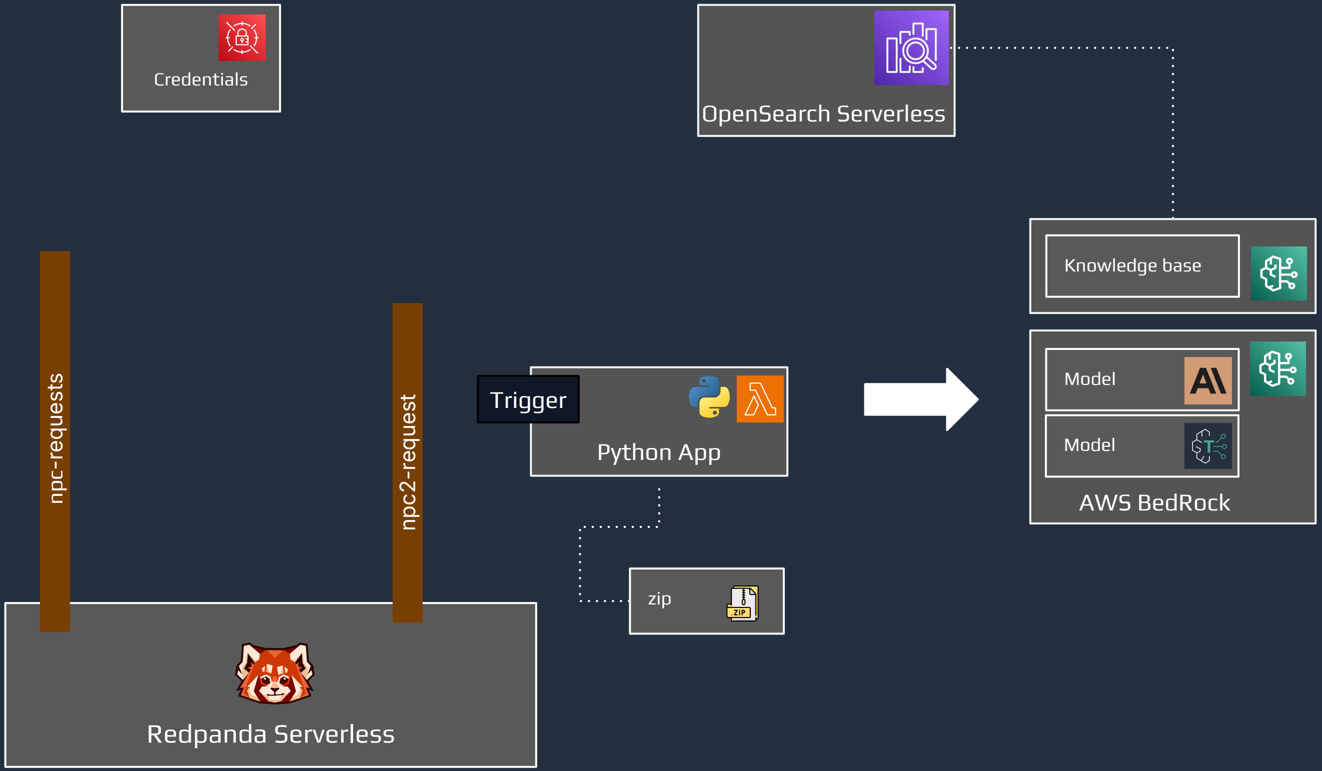
```yaml
processors:
    - ollama_chat:
        model: llama3.1
        prompt:
```

Output

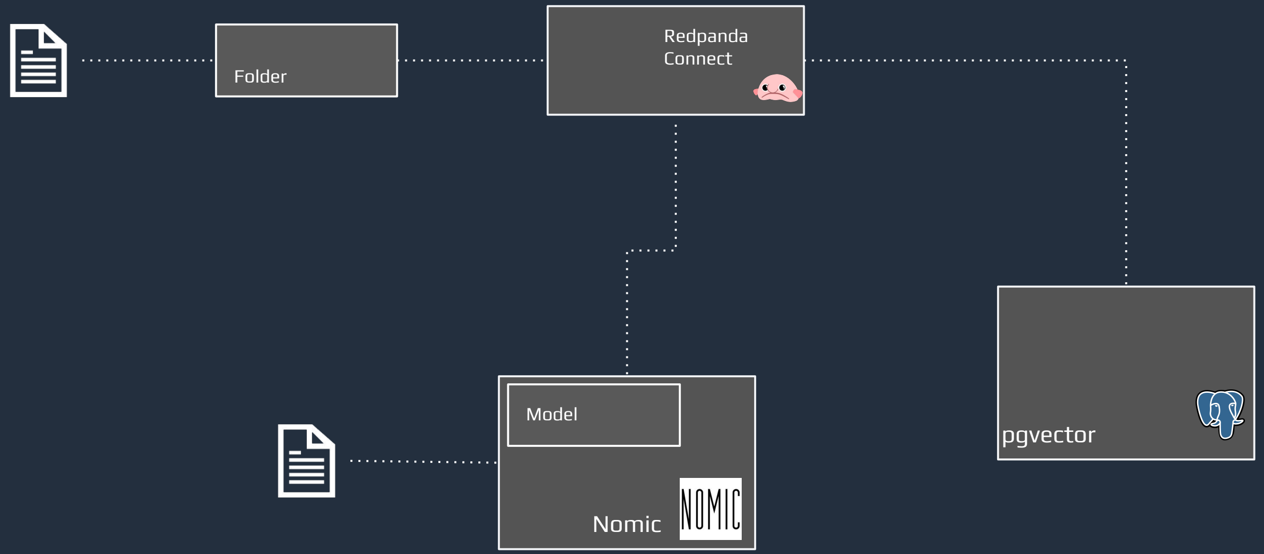
```yaml
output:
  kafka_franz:
    seed_brokers:
    topic: ""
```

Workshop

Components

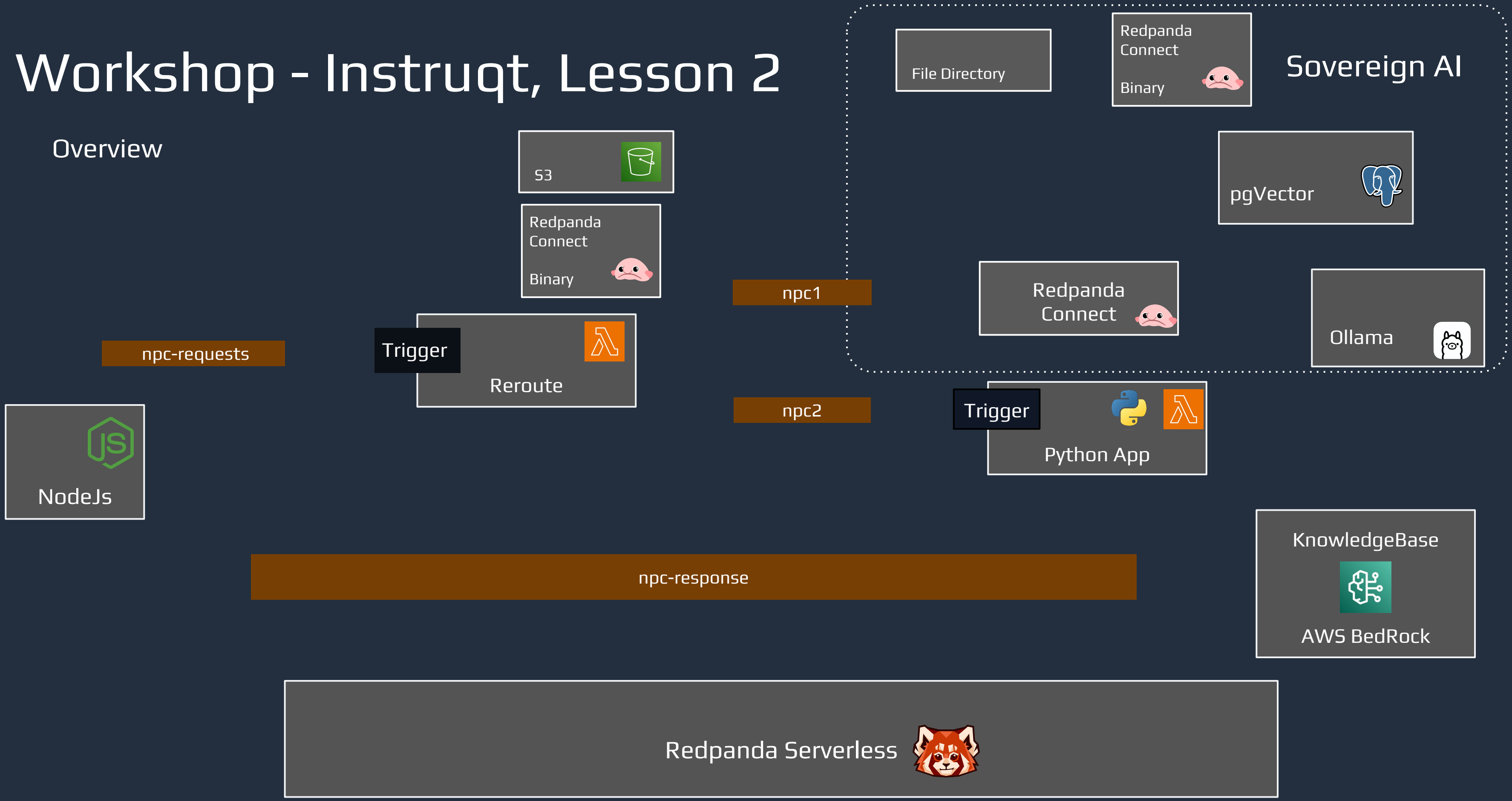


Workshop - Instruqt, Lesson 2



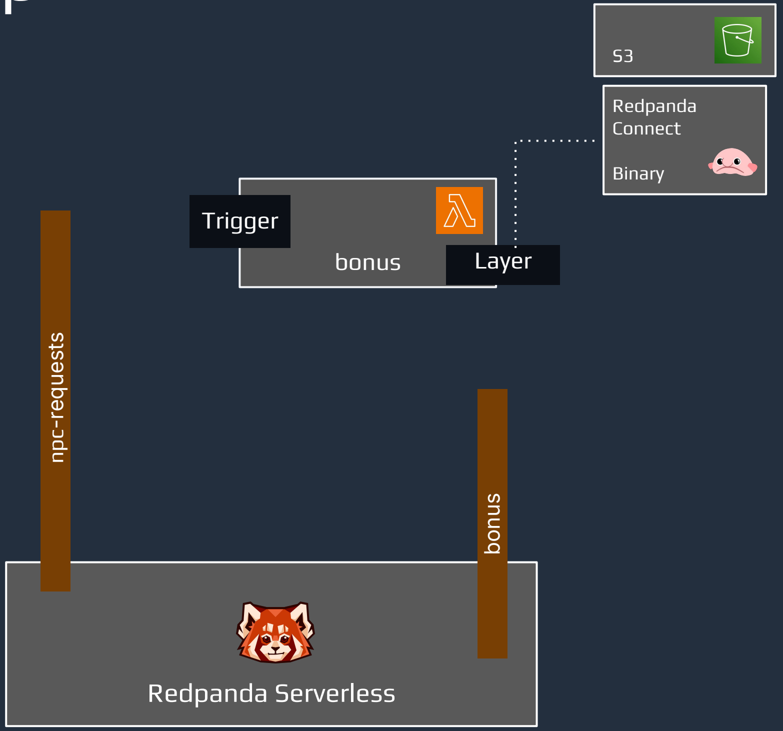
Workshop - Instruqt, Lesson 2

Overview



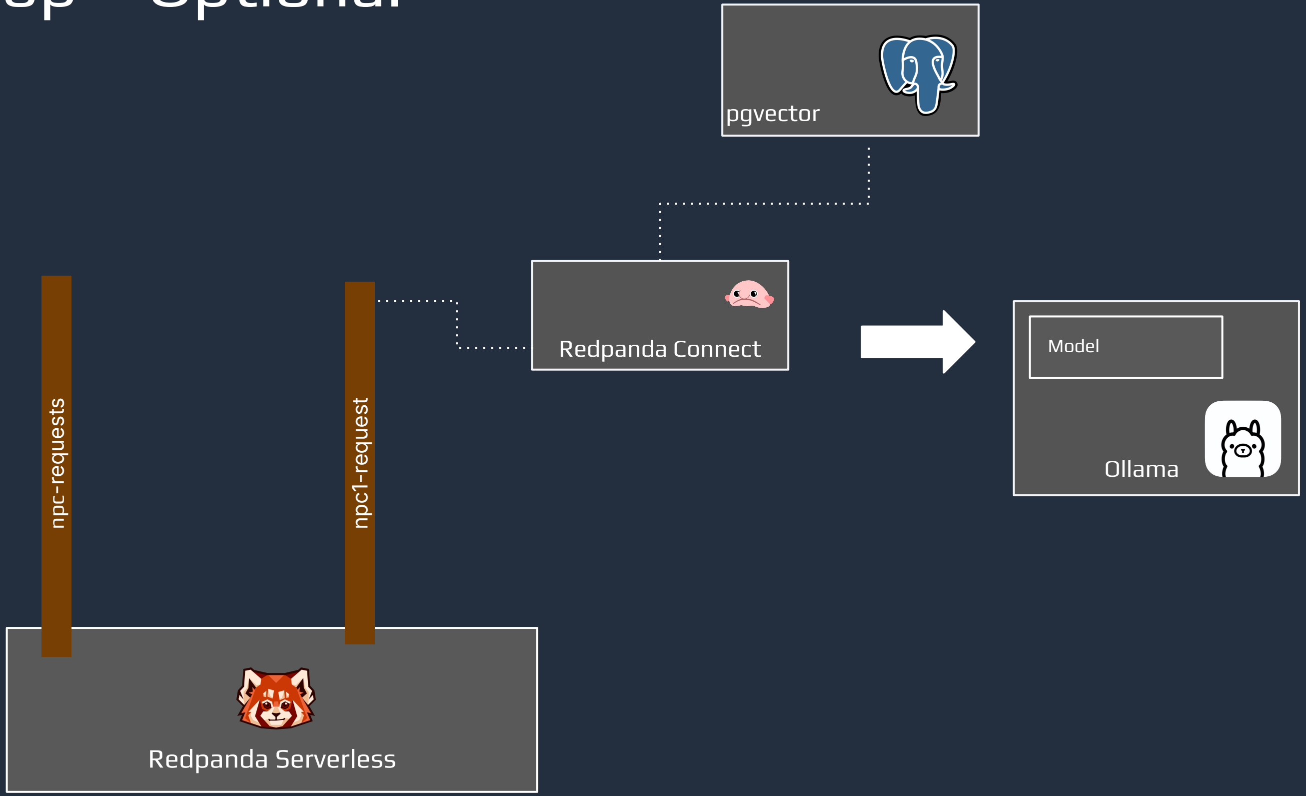
Workshop

Components



Workshop - Optional

Components



How would rate this workshop?

Thanks for joining!

Let's keep in touch

@redpandadata

redpanda-data

redpanda-data

hello@redpanda.com